13 Baryogenesis: Status after the P14 Audit

Summary. Earlier releases carried $\eta_B = 6.02 \times 10^{-10}$ as a derived output at [T2]. The P14 audit withdraws that claim. Equation (1) conserves psion number exactly, field-theory CP is exact, and the Standard-Model sphaleron conversion used in eq. (14) does not apply in the PH. Below Λ_{meta}, every mechanism tested is closed. The baryon asymmetry is therefore, at present, an initial condition, or is fixed at the condensation scale by physics that eq. (1) does not yet describe. What survives is structural: a set of conservation laws, a sharp statement of what the missing ingredient must do, and predictions that follow from the structure alone. One mechanism remains viable at the condensation scale, level bifurcation at formation (13.7). Its structure is confirmed by simulation, and its magnitude reduces to three quantities still to be derived. This section replaces 13.1–13.4; 13.5 is retained with a status note.

13.1 What Equation (1) Conserves

Psion number. N_ψ is the Noether charge of the overall phase and is exactly conserved by every term of eq. (1). In the broken phase it is the only exact additive charge (WP-D0: exact charge audit over 20 triples and four condensate variants, zero further candidates).

Continuous internal symmetry. The full Lie algebra of internal transformations leaving $(\bar{\psi}J^a\psi)(\bar{\psi}\Sigma^a\psi)$ invariant is two-dimensional and abelian, $U(1)_N \times U(1)_M$, with $M = \sum m_k$; no continuous chiral extension survives the G_{UV} term (WP-TOP). The winding condensate of App. J.6.2 breaks $U(1)_M$, leaving a residual discrete charge $D = (2M - 6Q + 3) \bmod 4$ (post-hoc observation, WP-TOP).

CP. The coupling tensor is real and ε_ψ is CP-even: field-theory CP is exact (WP-CP). The Z6 vacua break CP spontaneously, but conjugate vacua are degenerate ($\Delta C(s)$ symmetric).

Antimatter. Only one reading is consistent with observed reactions: antiparticles are composites of antipsions ($N_\psi = -3$, charge $-Q$). The index inversion $k \to (6 - k) \bmod 6$ of App. A.5 describes the Z6 label of the antipsion, not a psion composite. The alternative reading (antiparticle = psion triple with inverted indices) fails psion-number conservation already in $\beta^\pm$ decay, fails charge conservation in $n \to pe^-\bar{\nu}$ and $\mu^- \to e^-\bar{\nu}_e\nu_\mu$, and gives $\bar{\Sigma}^+$ the charge $+1$ (WP-ONT).

13.2 The Sakharov Conditions, Revisited

(1) Baryon-number violation. Nothing built from psions violates N_ψ, including an emergent sphaleron. With $N_\psi/3 = B + L$ (13.4), $B + L$ is conserved and only $B - L$ can change: the reverse of the Standard Model. The sphaleron factor $8/23$ and the coefficient C_{core} of the previous eq. (14) therefore have no basis in the PH. In addition, $C_{\text{core}} = 8.4434 \times 10^{-4}$ could not be reproduced from its stated inputs.

(2) C and CP violation. Eq. (1) supplies none (13.1). The former statement that the chirality-flip amplitude $\propto (m_e/\Lambda_{\rm meta})^2$ is a CP-odd phase is withdrawn: a real amplitude carries no phase. A phase inserted by hand, $J{\cdot}\Sigma \to {\rm Re}(e^{-i\varphi}O)$, is a rule outside eq. (1). With full density-matrix assembly kinetics it generates an asymmetry, and $\varphi \sim 10^{-4}$–10^{-2} suffices in magnitude. It does not, however, select the observed composites: exotic triples receive 2–2500 times the proton asymmetry, and Δe has the wrong sign in most runs (WP-LOOP).
(3) Departure from equilibrium. Available: assembly of composites near the contact-rate transition is out of equilibrium (WP-SIM, WP-NOW). Temperature histories alone redistribute abundances among cascade levels but never create an asymmetry (WP-TPROG).

13.3 Withdrawn and Reclassified Results

Withdrawn. (i) Eqs. (14)–(15), $\eta_B = C_{\rm core}(g_{1/2}/g_S)\varepsilon_\psi = 6.02\times10^{-10}$: no PH basis for the sphaleron input, and $C_{\rm core}$ is not reproducible. (ii) Eqs. (37)–(38) of App. J: $\xi^* = 434.09$ was computed from the target η_B, so eq. (38) is an identity, not a check. Independent routes to ξ^* give $\xi^* \le 1.08$ (Potts Z3, second-order NJL transition with $T_c = 9.6$ MeV) and $\xi^* = 3.6$–39 (Coleman–Weinberg with thermal inflation), none near 434. (iii) The claim that the PH satisfies all three Sakharov conditions without additional ingredients.
Reclassified. $\varepsilon_\psi = 25/169$ remains a result of $\mathrm{SU}(2)_{\rm meta}$ geometry (App. J, conditional on P23). Its link to η_B is not established and is moved from [T2] to open.
Closed mechanisms below $\Lambda_{\rm meta}$. Each was tested under a pre-registered definition, and each fails for a stated reason:

- Emergent sphalerons: they conserve N_ψ.
- Z6 domains with spontaneous CP: they form at $T \approx 3$ MeV, when assembly is frozen ($\Gamma/H = 3\times10^{-37}$). Their maximal comoving size today is ~ 30 pc, against the $\gtrsim 1$ Gpc required of antimatter regions. Degenerate conjugate vacua give a zero global average (WP-DOM).
- Temperature or pressure histories: no asymmetry from symmetric initial data (WP-TPROG, P14-D).
- Simple CP terms built from PH operators: 16 commutator and rearrangement patterns, alone or in pairs, fit the observed composition no better than random directions (WP-OPS).
- A phase in $J{\cdot}\Sigma$: no selection (WP-LOOP).
- Species mixing in the minimal extension of 13.4: the proton lifetime bounds its coefficient to $c \le 1.3\times10^{-8}$, which places the active window above 2.7×10^{15} GeV, outside the contact description (WP-ADD1).

Calibration is not derivation. A recipe exists that reproduces η_B, charge neutrality, the absence of exotic relics and $\Omega_{\rm DM}h^2 = 0.12$. It uses an explicit global CP violation of 5.7% in the assembly and a 69-component direction (WP-CAL). It is a reconstruction with no predictive weight: 69 parameters against 24 constraints always admit a solution, a factor 2 in the assembly rate reverses the sign of η_B, and 1% noise in the direction restores exotic relics at up to 33 times the baryon excess.

13.4 Baryon Number and the Level Rule

What observation requires. With the cascade level treated as a quantum number, 12 observed reactions (including $\beta^{\pm}$ decay, e^- capture, π, μ, Λ and Σ^0 decays, ν scattering and oscillations) require exactly two additive charges: B (states of the level-8 class) and L (level-10 and neutrino classes), with $N_\psi/3 = B + L$ (WP-LEV1).

What eq. (1) allows. With only N_ψ and Q conserved, seven unobserved processes are allowed: $p \to \nu\pi^+$, $n \to \nu\pi^0$, $n \to 3\nu$, $p\,e^- \to \nu_\mu\nu_\tau$, and three level jumps within one triple ($n \to \nu_\tau\gamma$, $\Lambda^0 \to \nu_e\gamma$, $\Sigma^0 \to \nu_\mu\gamma$). All have $\Delta B = -\Delta L$. Against the measured limits ($p \to \bar\nu\pi^+ > 3.9 \times 10^{32}$ yr, $n \to \bar\nu\pi^0 > 1.1 \times 10^{33}$ yr, $n \to$ invisible $> 5.8 \times 10^{29}$ yr), the required amplitude suppression is $\lesssim 10^{-8}$ relative to a contact operator at $\Lambda_{\rm meta}$. For same-triple jumps it is $\lesssim 5 \times 10^{-30}$ relative to an electromagnetic transition (reference lifetime 10^{29} yr) (WP-LEV1).

What eq. (1) cannot supply. B is not a Noether charge: it lies outside the span of (M, Q, N_ψ). It is not topological: the vacuum manifold has $\pi_2 = \pi_3 = 0$, so no skyrmion-like baryon number exists. The level is not an internal degree of freedom: states are labelled by the triple alone (WP-TOP). Hence the dual-sector identification of §6 and L.4.4, in which the same triple is either a baryon or a neutrino ((0,1,5) $= n/\nu_\tau$, (0,2,4) $= \Sigma^0/\nu_\mu$, (1,2,3) $= \Lambda^0/\nu_e$), is incompatible with neutron stability as eq. (1) stands. Correction to §6: there are five neutral three-body triples, not three. They are (0,1,4), (0,1,5), (0,2,4), (0,2,5) and (1,2,3); the two additional ones carry $D = 2$.

Minimal extension (registered, not adopted). A two-valued species index, $\psi \to \psi_s$ with $s \in \{B, L\}$ and the currents of eq. (1) summed over species, preserves $U(1)_B \times U(1)_L$ if $SU(2)_{\rm species}$ is broken (for example, $G_{{\rm UV},B} \neq G_{{\rm UV},L}$). All 12 observed reactions are allowed, all 7 unobserved processes are forbidden exactly, and the level becomes a property of the species. The costs (WP-ADD1):

- The L sector contains six positively charged matter leptons, (0,4,5), (1,3,4), (1,3,5), (2,3,4), (2,3,5) and (3,4,5), which must be heavy.
- The numerical predictions survive only if the species index does not enter $|\beta_0| = N_{\rm Dirac} + 2/N_f$. Doubling N_f gives $m_e = 0.065$ MeV; doubling $N_{\rm Dirac}$ gives $m_e \times 6 \times 10^{10}$.

A non-running species label is equivalent to postulating baryon-number conservation. This is recorded as open problem **P14-B**: derive a conserved baryon label from the condensate, or adopt one and re-derive the counting theorems.

13.5 Link to N_{eff} (status note)

Equation (16) is retained as written, but its shared-parameter status with η_B lapses with 13.3. If $\mu_\nu/T = 0.123$ is read as a lepton-number asymmetry, it conflicts with psion-number bookkeeping. With total $N_\psi = 0$, $B = -L$ and charge neutrality, the forced antineutrino excess is two per hydrogen atom, which implies a degeneracy $\xi \approx 2.5 \times 10^{-9}$, some 5×10^7 times smaller than 0.123 (WP-CAL, V3). Equation (16) must therefore rest on a different reading or be reclassified.

13.6 Structural Predictions that Survive

(a) No nucleon decay with $\Delta B = \Delta L$: $p \to e^+\pi^0$ and similar modes change N_ψ and are forbidden exactly [T1, from eq. (1)]. **(b)** Any nucleon decay, if observed, has $\Delta B = -\Delta L$ ($p \to \nu\pi^+$, $n \to \nu\pi^0$, $n \to 3\nu$). **(c)** No $0\nu\beta\beta$ and no n–$\bar{n}$ oscillation, since both change N_ψ. **(d)** If the baryon asymmetry was set with total $N_\psi = 0$: a cosmic antineutrino excess of two per hydrogen atom, at degeneracy $\xi \approx 2.5 \times 10^{-9}$. **(e)** Dark matter as a thermal relic of the cascade top requires level ≈ 4.8 (≈ 56 TeV, $\kappa = 4\pi$). The canonical $m_{\text{DM}} = 7.2 \times 10^{10}$ GeV would overclose by $\sim 10^{11}$–10^{12} if thermal. This bears on §14.3, whose direct route also depends on ξ^* (13.3).

13.7 A Viable Mechanism: Level Bifurcation at Formation

Hypothesis. The dual sector of §6, a liability in 13.4, supplies what every earlier test lacked: two final channels from one initial state. A psion triple formed near Λ_{meta} binds into either the baryonic rung (t_B) or the leptonic rung (t_L). Two channels from a common initial state is the minimal structure in which CP violation is compatible with CPT and unitarity. The hypothesis (WP-BIF) has three parts. (i) Formation is biased between the channels by ε, with opposite sign for antipsion triples. (ii) The CP source is ε times the net formation rate, the leptogenesis form, so it vanishes in equilibrium; dissociation followed by re-formation washes the channel imbalance. (iii) Once formed, the rung is conserved. Part (iii) is the label of P14-B; here it is postulated, not derived.

What follows from the structure. The following come out of the dynamics without being imposed (pre-registered network with 92 species: psions, antipsions and the four states $t_B, t_L, \bar{t}_B, \bar{t}_L$ of each triple).

- $B = -L$ exactly: $|B+L|/|B| \leq 6 \times 10^{-10}$ wherever $|B|$ is above numerical noise.
- Total charge zero and N_ψ conserved, both to $\sim 10^{-13}$.
- No asymmetry in equilibrium, even with $\varepsilon = 0.1$.
- Linearity in ε to 3×10^{-8}, and exact reversal under $\varepsilon \to -\varepsilon$.

With B, L and Q conserved and p, e, ν the lightest states of their sectors, the only endpoint is hydrogen plus two net antineutrinos per baryon. Exotic triples decay into it by energetics, so the composition problem of WP-CAL (13.3) does not arise: it reflected the absence of a conserved label, not a missing tuning.

Magnitude. The efficiency η_B/ε is computed with dilution $D = 1$; network normalisation as in WP-SIM. With a universal bias it peaks at 3.6×10^{-3} near the contact-rate transition ($c_\Gamma = 10^{-3}$). Over $c_\Gamma = 10^{-5}$–10^{-2} it varies by a factor ~ 20 with a stable sign, and it is washed out for $c_\Gamma \geq 0.1$. The result is therefore not fine-tuned, unlike the calibration of 13.3. A bias weighted by triple chirality $(n_L - n_R)/3$ changes sign between binding laws and cannot fix the sign. The observed η_B requires

$$\varepsilon\, D \simeq 6.1 \times 10^{-10} \,/\, (\eta_B/\varepsilon) \;\gtrsim\; 1.7 \times 10^{-7}.$$

Post-hoc reading (declared as such): if $|\varepsilon| = \varepsilon_\psi = 25/169$ and $D = \xi^{*-3}$, the observed value requires $\xi^* \approx 36$–95. The Coleman–Weinberg route with thermal inflation gives $\xi^* = 3.6$–39; the ranges touch but do not convincingly overlap.

Status and conditions. [T3]: the mechanism is identified and its structure verified, but its rates rest on modelled contact kinetics (binding weights L1/L2, rate normalisation c_Γ), so the magnitude is plausible, not predicted. Promotion requires three derivations:

(C1) The conserved rung label (P14-B). The same principle must secure proton stability (13.4) and enable this mechanism.

(C2) The bias ε and a global sign. ε_ψ is CP-even and can at most set the size; the sign must come from a vacuum orientation shared by the whole observable universe, the natural candidate being the winding direction of the condensate (App. J.6).

(C3) The departure from equilibrium (c_Γ) and the dilution ξ^*, both from the transition dynamics.

If C1–C3 close with η_B in range, the PH would obtain baryogenesis and nucleon stability from a single principle. Its characteristic signature is the cosmic antineutrino excess of 13.6(d).

L.6 P14 Audit Record (October 2026)

All work packages were pre-registered, with SHA-256 and a UTC timestamp, before execution. Failures and corrections are recorded with the same weight as results; the full log is in `p14d_psion_number_audit/README.md` of the companion repository [2].

L.6.1 Errors made during the audit and corrected. (i) Sphalerons were first treated as external to the PH; they are psion composites and conserve N_ψ. (ii) Antimatter was first modelled as antipsions, then switched to index-inverted psion triples on a reading of A.5, and restored to antipsions by WP-ONT; the second step was the wrong one. (iii) WP-DOM, variant V2: unitarity was first

imposed jointly across processes with different density scaling, which created asymmetry in equilibrium. Control C4 caught this; the variant was corrected to per-initial-state unitarity, and the faulty version is retained. (iv) WP-CAL: the dark-matter block first patched a copy of the module globals, so the coupling change never reached the simulation; it was redone in an addendum. (v) WP-TOP: the anticipated count of three neutral triples was wrong (it is five). (vi) WP-BIF: the registered endpoint bookkeeping had a sign error ($N_{\bar{\nu}} - N_{\nu} = -L - N_e$ instead of $N_e - L$), repeated by the script, which printed zero; the corrected count is two per baryon. The anticipated efficiency (10^{-6}–10^{-5}) was wrong by about three orders of magnitude.

L.6.2 Registered control failures. WP-NOW C1/C2 (residual 2×10^{-13} against 10^{-15}, floating-point floor); WP-TPROG C3 (first-order step error of 1–2%); WP-DOM C3 (2.7×10^{-6}) and C4 (1.3×10^{-12}, floor); WP-LOOP C3 (1.8×10^{-6}), and its $c_\Gamma = 1$ sensitivity run not completed (explicit integrator); WP-BIF C3 ($B + L = 0$ fails formally for $c_\Gamma \geq 0.1$, where $|B| \sim 10^{-14}$ is at the numerical floor).

L.6.3 Work packages. WP-B, WP-C, WP-A1, WP-A2 (P14 core); P14-D, WP-D0, WP-D0′, WP-CP, WP-J1, WP-M, WP-SIM, WP-ONT, WP-NOW, WP-TPROG, WP-DOM, WP-CAL, WP-OPS, WP-LOOP, WP-LEV1, WP-TOP, WP-ADD1, WP-BIF. Interactive simulators: `simulador/`.